Finally, instructors saw great potential for such a system that can help them reflect on and improve their teaching. Reviewing patterns in student confusion could help adjust explanations and lecture pacing. As I6 stated, "*If I've got the most common types of questions for a particular lecture, I can use that information when I plan that lecture for the next time.*"

Design Implication #5: These insights suggest that an effective AI-powered classroom assistant should not only support real-time clarification for students, but also provide actionable feedback to instructors to enhance teaching effectiveness.

4 AskNow

Based on the formative study findings, we designed and developed AskNow, an interactive system powered by a large language model (LLM) to support real-time question answering in large-scale classrooms. AskNow consisted of a student and an instructor interfaces. We show the architecture of AskNow in Figure 2.

We carefully structured AskNow's functionality based on the explicit challenges and requirements identified from our formative study. In particular, each core feature directly addresses the five Design Implications (DIs #1–#5) we derived in Section 3. Students reported significant difficulties in following fast-paced lectures, language and cultural barriers causing comprehension issues, and hesitancy in public questioning due to social anxiety. Conversely, instructors expressed challenges in gauging real-time student comprehension, and wanted effective tools to detect and address areas of widespread confusion promptly. In response, AskNow incorporates three core phases to directly address these identified needs: (1) real-time lecture transcription, (2) context-aware question answering, and (3) question aggregation.

4.1 Real-time Lecture Transcription

On the instructor interface, AskNow provides real-time, automatic speech-to-text transcription of lecture audio. This real-time transcription functionality not only provides instructors with a comprehensive textual record of their lecture but also creates an essential context data for the language model, serving subsequent functionalities such as context-aware question answering. This ensures that AskNow-generated responses are precise, contextually relevant, and directly linked to the ongoing lecture content, which directly addresses Design Implication #1 by offering students immediate clarifications to help them catch up on missed content and supports Design Implication #3 through continuous real-time availability of contextual transcript information.

4.2 Context-aware Question Answering

On the student interface, AskNow allows students to privately submit questions and obtain real-time, contextually accurate responses,

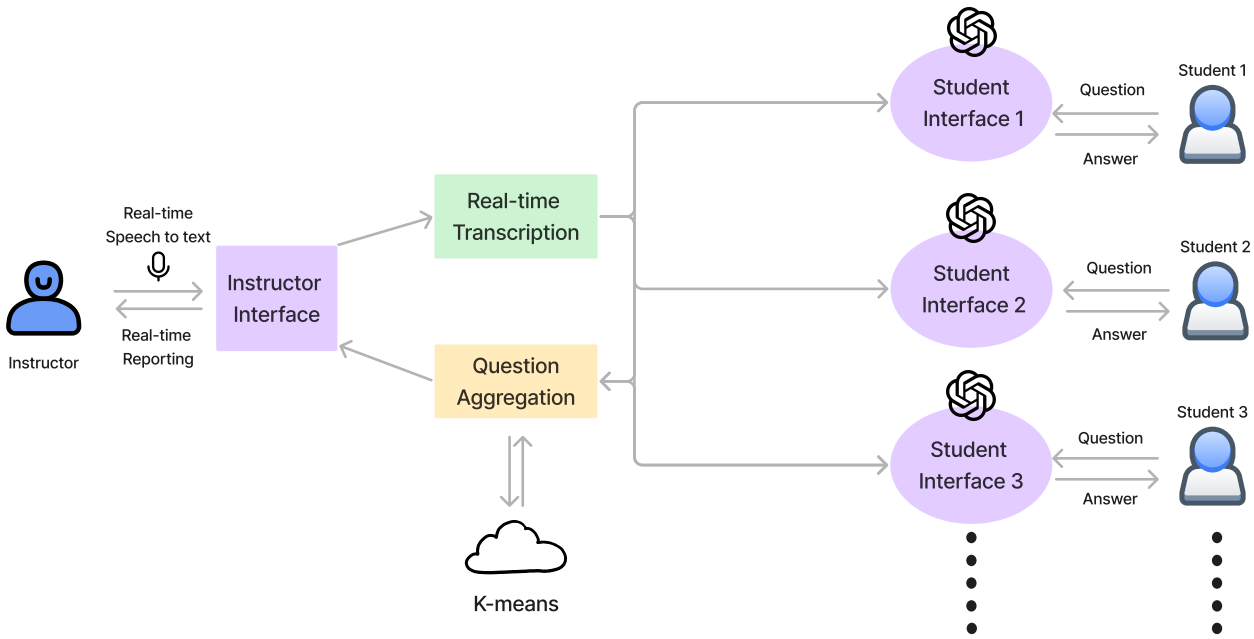


Figure 2: AskNow system architecture and end-to-end data flow. Student interfaces send questions to a cloud backend, which retrieves the ongoing lecture transcript (streamed from the Instructor Interface via real-time speech-to-text) and lecture metadata to construct a context-aware prompt for the LLM; the generated answer is then returned to the originating student. In parallel, all question events are logged, embedded, and clustered into topic groups that are displayed on the Instructor Interface to surface common areas of confusion in real time.